

St George Girls High School

Year 12

Assessment Task 3

2007



Mathematics

Extension 1

General Instructions

- Time allowed – 75 minutes
- Write using blue or black pen
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.
- Write on one side of the page only.
- Start each question on a new page.

Total marks – 70

- Attempt Questions 1 – 5
- All questions are of equal value

Question	Mark
Question 1	/14
Question 2	/14
Question 3	/14
Question 4	/14
Question 5	/14
Total	/70

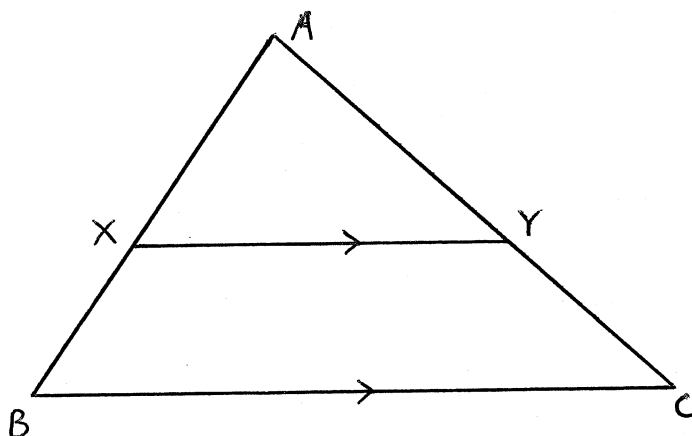
Question 1 – (14 marks) – Start a new page

Marks

a) Differentiate $x \tan^{-1} \frac{x}{2}$

3

b)



In the triangle ABC , $XY = 8\text{cm}$, $BC = 14\text{cm}$, $AC = 18\text{cm}$ and $XY \parallel BC$.

(i) Prove that $\triangle AXY$ is similar to $\triangle ABC$

3

(ii) Find the length of AY giving reasons.

3

c) For the function $y = \sin^{-1}\left(\frac{x}{2}\right)$:

(i) State the domain and range

2

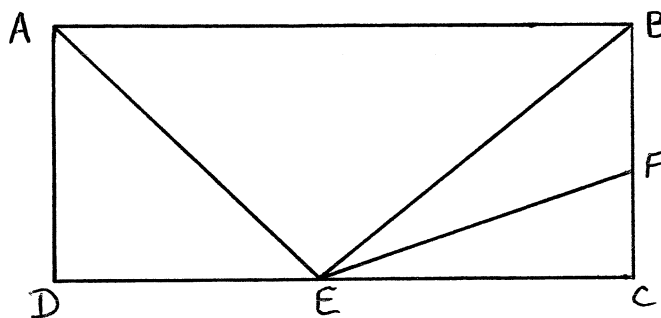
(ii) Sketch the graph of the function

3

Question 2 – (14 marks) – Start a new page

Marks

a)



$ABCD$ is a rectangle. E is a point on DC such that AE bisects angle DEB and EF bisects angle BEC . Prove that angle $AEF = 90^\circ$

3

b) Evaluate $\int_0^1 \frac{1}{\sqrt{4-3x^2}} dx$

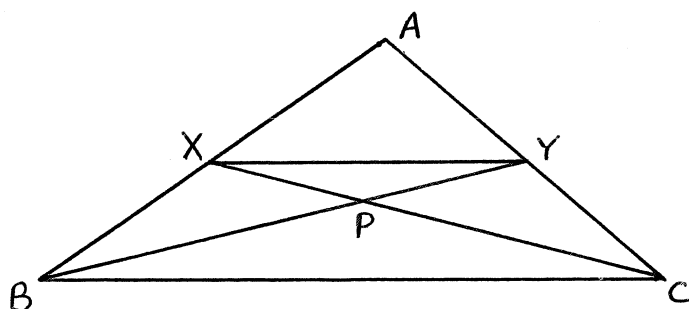
2

c) Find the value of the expression

$\cos^{-1}\left(-\frac{1}{2}\right) - \sin^{-1}\left(-\frac{1}{2}\right)$ in terms of π

2

d) In the diagram below, $\angle AXY = \angle AYX$ and $XP = YP$



(i) Copy this diagram on your sheet.

1

(ii) Prove that $\triangle ABY \equiv \triangle ACX$, giving reasons.

4

(iii) Hence prove that $\triangle BPC$ is isosceles.

2

Question 3 – (14 marks) – Start a new page

Marks

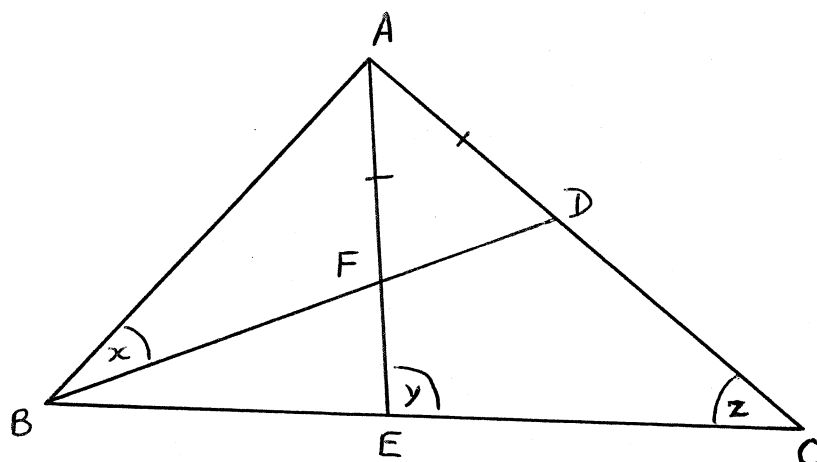
a) (i) If $\theta = \tan^{-1} A + \tan^{-1} B$ show that $\tan \theta = \frac{A+B}{1-AB}$

1

(ii) Hence solve the equation $\tan^{-1} 3x + \tan^{-1} 2x = \frac{\pi}{4}$

4

b)



The diagram shows triangle ABC . The bisector of angle B meets the line AE at F and the line AC at D .

If $\angle ABD = x^\circ$

$\angle AEC = y^\circ$

$\angle ACB = z^\circ$

and $AD = AF$ show that

(i) $\angle ADF = \frac{1}{2}(y+z)$

3

(ii) $\angle EAB = y - 2x$

2

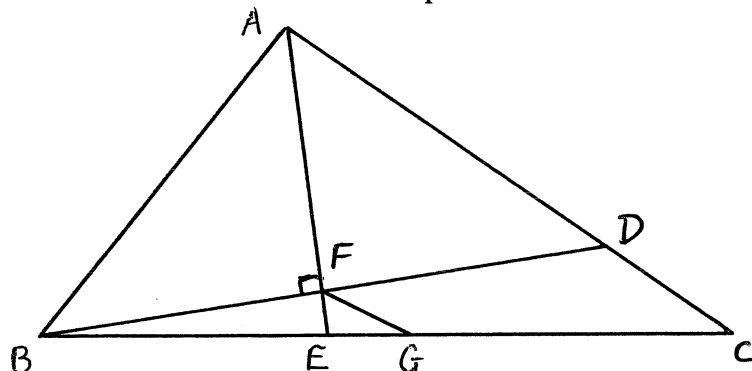
c) Find the equation of the tangent to the curve $y = 3\cos^{-1} \frac{x}{2}$ at the point on the curve where $x = 0$

4

Question 4 – (14 marks) – Start a new page

Marks

- a) In the diagram AE bisects $\angle BAC$, BF is perpendicular to AE and G is the midpoint of BC . BF meets AC at the point D .

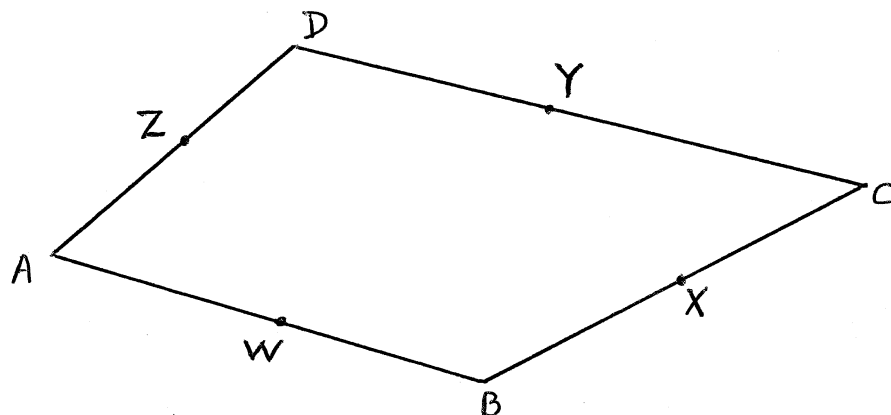


- (i) Copy this diagram onto your answer sheet and mark in all the given information. 1
- (ii) Prove that $\triangle BAF$ is congruent to $\triangle DAF$ 3
- (iii) Explain why $BF = FD$ 1
- (iv) Hence prove that FG is parallel to DC 4
- b) (i) Find $\frac{d}{dx} \left(\sqrt{1-x^2} + x \sin^{-1} x \right)$ 3
- (ii) Hence or otherwise evaluate $\int_0^{\frac{1}{2}} \sin^{-1} x \, dx$ correct to 3 significant figures. 2

Question 5 – (14 marks) – Start a new page

Marks

a)



$ABCD$ is a quadrilateral. If W is the midpoint of AB , X is the midpoint of BC , Y is the midpoint of CD and Z is the midpoint of AD , prove that the quadrilateral $WXYZ$ is a parallelogram.

b) Consider the function $f(x) = \frac{8}{4+x^2}$

(i) Show that $f(x)$ is an even function, and the x axis is a horizontal asymptote to the curve $y = f(x)$

(ii) Find the coordinates and nature of the stationary point on the curve $y = f(x)$

(iii) Sketch the graph of the curve showing the above features.

(iv) Find the exact area of the region in the first quadrant bounded by the curve $y = f(x)$ and the line $x = 2$

Question 1

a) $\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$
 $u = x \quad v = \tan^{-1} \frac{x}{2}$
 $\frac{du}{dx} = 1 \quad \frac{dv}{dx} = \frac{2}{4-x^2}$
 $\frac{dy}{dx} = \tan^{-1} \frac{x}{2} + \frac{2x}{4-x^2}$

b) (i) For $\triangle AXY$ and $\triangle ABC$

$\angle A$ is common

$\angle AXY = \angle ABC$ (corresponding angles
on $XY \parallel BC$)

$\therefore \triangle AXY \parallel \triangle ABC$ (equiangular)

(ii) $XY : BC = 8 : 14$ sides of similar
 $= 4 : 7$ triangles

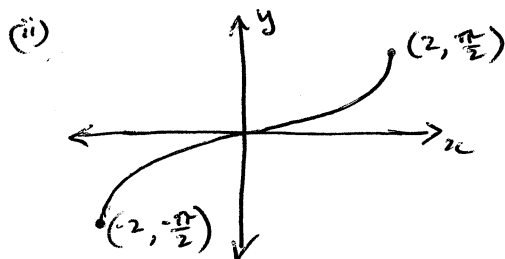
$AY : AC = 4 : 7$ "

$$\frac{AY}{AC} = \frac{4}{7}$$

$$\frac{AY}{18} = \frac{4}{7}$$

$$AY = \frac{72}{7} \text{ cm}$$

c) (i) Domain $-2 \leq x \leq 2$ range $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$



Question 2

a) Let $\angle AED = x$

$\therefore \angle AEB = x$

Let $\angle CEF = y$

$\therefore \angle BEF = y$

$x + x + y + y = 180^\circ$ angles on
a straight line

$$2x + 2y = 180$$

$$x + y = 90$$

$$\angle AEF = x + y$$

$$= 90^\circ$$

b) $\int_0^1 \frac{1}{\sqrt{4-3x^2}} dx = \left[\frac{1}{\sqrt{3}} \sin^{-1} \frac{\sqrt{3}x}{2} \right]$
 $= \frac{1}{\sqrt{3}} \left(\frac{\pi}{3} - 0 \right)$
 $= \frac{\pi}{3\sqrt{3}}$

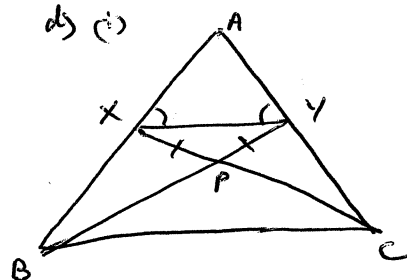
c) $\cos^{-1} \left(-\frac{1}{2} \right) = \frac{2\pi}{3}$

$$\sin^{-1} \left(-\frac{1}{2} \right) = -\frac{\pi}{6}$$

$$\cos^{-1} \left(-\frac{1}{2} \right) - \sin^{-1} \left(-\frac{1}{2} \right) = \frac{2\pi}{3} - \left(-\frac{\pi}{6} \right)$$

$$= \frac{5\pi}{6}$$

d) (i)



(ii) For $\triangle ABY$ and $\triangle ACX$

$\angle PXY = \angle PYX$ angles opposite
equal sides

$\therefore \angle AXC = \angle AYB$ sums of
equal angles

$\angle A$ is common

$AX = AY$ sides opposite
equal angles

$\therefore \triangle ABY \equiv \triangle ACX$ (AAS)

(iii) $BY = CX$ (corresponding sides
of congruent $\triangle$'s)

$$BY - PY = CX - PX$$

$$BP = CP$$

$\therefore \triangle BPC$ is an isosceles $\triangle$

Question 3

$$\begin{aligned} a) (i) \tan \theta &= \tan(\tan^{-1} A + \tan^{-1} B) \\ &= \frac{\tan(\tan^{-1} A) + \tan(\tan^{-1} B)}{1 - \tan(\tan^{-1} A) \tan(\tan^{-1} B)} \\ &= \frac{A + B}{1 - AB} \end{aligned}$$

$$\begin{aligned} (ii) \quad A &= 3x \quad B = 2x \\ \tan^{-1} 3x + \tan^{-1} 2x &= \frac{3x + 2x}{1 - 3x \times 2x} \\ &= \frac{5x}{1 - 6x^2} \end{aligned}$$

$$\frac{5x}{1 - 6x^2} = \tan \frac{\pi}{4}$$

$$\frac{5x}{1 - 6x^2} = 1$$

$$5x = 1 - 6x^2$$

$$6x^2 + 5x - 1 = 0$$

$$6x^2 + 6x - x - 1 = 0$$

$$6x(x+1) - (x+1) = 0$$

$$(x+1)(6x-1) = 0$$

$$x = -1 \text{ or } \frac{1}{6}$$

$$b) (i) \angle AFD = \angle ADF \text{ angles opposite equal sides}$$

$$\angle EAC = 180 - (y + z)$$

$$\angle AFD + \angle ADF = y + z$$

$$\therefore \angle ADF = \frac{y+z}{2}$$

$$(ii) \angle ABE + \angle BAE = \angle AEC \text{ (exterior angle of a triangle)}$$

$$2x + \angle EAB = y$$

$$\angle EAB = y - 2x$$

$$c) \frac{dy}{dx} = \frac{-3}{\sqrt{4-x^2}}$$

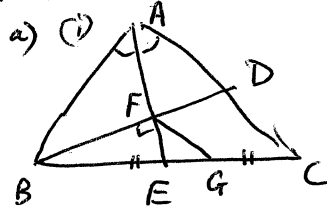
$$= -\frac{3}{2} \text{ when } x = 0$$

$$(0, \frac{3\pi}{2})$$

$$y - \frac{3\pi}{2} = -\frac{3}{2}x$$

$$3x + 2y - 3\pi = 0$$

Question 4



$$(ii) \text{ For } \triangle ABF \text{ and } \triangle DAF$$

$$\angle BAF = \angle DAF \text{ given}$$

$$\angle BFA = \angle DFA \text{ given as perpendicular}$$

$$AF \text{ is common}$$

$$\therefore \triangle ABF \cong \triangle ADF \text{ (AAS)}$$

$$(iii) BF = FD \text{ corresponding sides of congruent triangles}$$

$$(iv) \text{ For } \triangle BEF \text{ and } \triangle BCD$$

$$BF = FD \text{ above}$$

$$BG = GC \text{ given}$$

$$\angle B \text{ is common}$$

$$\therefore \triangle BEF \parallel \triangle BCD \text{ two sides in proportion and the included angle equal}$$

$$\therefore \angle BGF = \angle BCD \text{ corresponding angles of similar triangles}$$

$$\therefore FG \parallel DC \text{ (corresponding angles are equal on parallel lines)}$$

$$b) (i) \frac{d}{dx} (\sqrt{1-x^2} + x \sin^{-1} x)$$

$$= -\frac{1}{2} \frac{2x}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x$$

$$= -\frac{x}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x$$

$$= \sin^{-1} x$$

$$(ii) \int_0^{\frac{1}{2}} \sin^{-1} x \, dx$$

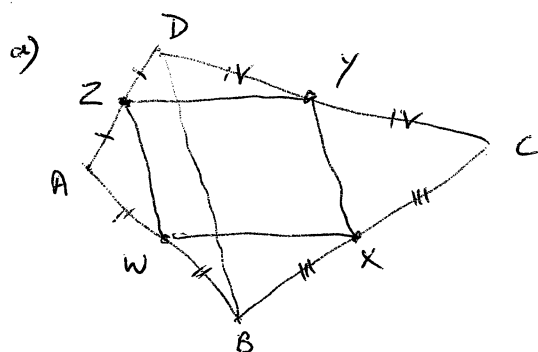
$$= \left[\sqrt{1-x^2} + x \sin^{-1} x \right]_0^{\frac{1}{2}}$$

$$= \left[\sqrt{1-\frac{1}{4}} + \frac{1}{2} \sin^{-1} \frac{1}{2} - 1 - 0 \right]$$

$$= \frac{\sqrt{3}}{2} + \frac{\pi}{12} - 1$$

$$= 0.128$$

Question 5



join BD

For $\triangle AWZ$ and $\triangle ABD$

$$AZ = ZD$$

$$AW = WB$$

$\angle A$ is common

$\therefore \triangle AWZ \parallel \triangle ABD$

two sides in proportion
and the included angles equal

$\therefore WZ \parallel BD$ equal intercepts

For $\triangle CXY$ and $\triangle CBD$

$$CX = XB$$

$$CY = YD$$

$\angle C$ is common

$\therefore \triangle CXY \parallel \triangle CBD$

two sides in proportion
and the included angles equal

$\therefore XY \parallel BD$ equal intercepts

$\therefore WZ \parallel XY$

join AC

and use similar proof of

$$WX \parallel ZY$$

$\therefore WXYZ$ is a parallelogram

$$\begin{aligned} b) (i) f(-x) &= \frac{8}{4+(-x)^2} \\ &= \frac{8}{4+x^2} \\ &= f(x) \end{aligned}$$

$\therefore$ it is an even function

$$(ii) f'(x) = \frac{-16x}{(4+x^2)^2}$$

$$f'(x) = 0 \text{ when } x = 0$$

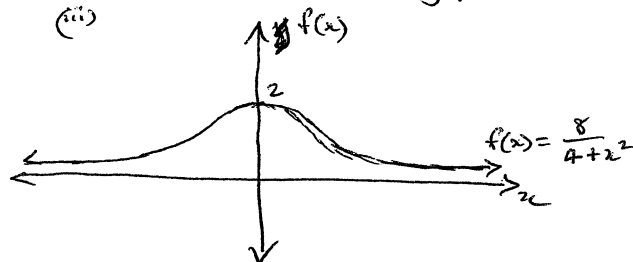
$$f(0) = 2$$

x	0^-	0	0^+
$f'(x)$	$+$	0	$-ve$



$\therefore$ a maximum turning pt at $(0, 2)$

(iii)



$$\begin{aligned} (iv) \int_0^{\infty} \frac{8}{4+x^2} dx &= \left[4 \tan^{-1} \frac{x}{2} \right]_0^{\infty} \\ &= 4 \tan^{-1} \infty - 4 \tan^{-1} 0 \\ &= \pi - 0 \\ &= \pi \end{aligned}$$

$$\begin{aligned} f(x) &= \frac{8}{x^2} \\ \lim_{x \rightarrow \infty} \frac{8}{x^2} &= \frac{0}{0+1} \\ &= 0 \end{aligned}$$

$\therefore$ horizontal asymptote at $f(x) = 0$